Total No. of Questions : 4]

SEAT No. :

PE-560

[Total No. of Pages : 2

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S.E. (Artificial Intelligence & Machine Learning) (Insem)

DATA STRUCTURES & ALGORITHMS

(2019 Pattern) (Semester - III) (218542)

Time : 1 Hour] *[Max. Marks : 30*

Instructions to the candidates:

1) *Attempt questions Q.1 or Q.2, Q.3 or Q.4*
2) *Figures to the right indicate full marks.*
3) *Draw neat & labelled diagram if necessary.*
4) *Assume suitable data if necessary.*

Q1) a) For a Singly Linked list, write a pseudocode to delete: **[5]**

i) First element

ii) Last element

b) What is an ADT? Explain ADT operations for singly Linked List. **[5]**

c) Explain frequency count method with the help of suitable example. **[5]**

OR

Q2) a) Discuss the- concept of circular linked list and doubly linked list with the structure of node and sample linked list. **[6]**

b) An array arr[10][5] is stored in the memory with each element requiring 2 bytes of storage. If the first element arr[0][0] is stored at the location 1250, calculate the address of arr[5][6] when the array is stored using row major technique. **[5]**

c) Calculate time complexity of following codes: **[4]**

i)
```
int a = 0,b = 0
  for (i = 0; i <N; i++)
{
     a = a + rand ( );
}
for (j = 0;j < M; j++)
{
   b = b + rand ( ) ;
}
```

P.T.O.

ii)
```
a = 0
i = n
While (i >0)
{
    a = a+i;
    i = i/2;
}
```

Q3) a) Explain quick sort & sort the given list using quick sort: **[6]**
15, 08, 20, -4, 16, 02, 01, 12, 21, -2

Analyse time complexity of quick sort.

b) Apply binary search on {10,20,30,35,40,45,50}' to-search the position of the element 40. Comment on it's time complexity. **[5]**

c) Differentiate between binary search and Fibonacci search. **[4]**

OR

Q4) a) Sort the following numbers using insertion sort: **[6]**

55, 85, 45, 11, 34, 05, 89, 99, 67, 55

Comment on efficiency, stability, in-place characteristics of insertion sort.

b) Suppose we have the array: (1, 2, 3,4, 5, 6,7). We have to find the element X = 6 using Fibonacci search. Explain all steps. **[5]**

c) Explain the stable and unstable sort with the help of suitable example. **[4]**

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